



In situ formed Co clusters in selective oxidation of α -C–H bond: Stabilizing effect from reactants

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ABSTRACT

Aerobic oxidation of α -C–H bond of organic compounds to valuable chemicals is widely investigated in both fundamental research and industry. Due to the good stability of molecular oxygen, severe reaction conditions are generally required. Herein, by *in situ* synthesis we used molecular oxygen to induce cobalt nanoclusters with the sensitive catalysis in mild selective oxidation. The cobalt containing clusters with an average diameter around 0.9 nm are *in situ* prepared in the presence of *cis*-cyclooctene epoxidation and cyclooctene dimer oxide is formed at the interface to stabilize Co clusters with electron donation as an oil-soluble surfactant. The soluble clusters exhibit high activity in selective oxidation of α -C–H bond of ethylbenzene into acetophenone and turnover number (TON) reaches about 7×10^4 during 50 h' reaction at 373 K, which is around 960 times more active than the one using CoCl_2 salt as the catalyst, resulting from efficient mass transportation, π bond interaction and oxygen gas activation. Extended work based on this understanding demonstrates that cobalt nanoclusters also effectively catalyze aerobic oxidation of cyclohexene.

1. Introduction

Green oxidation of aromatic alkyl C–H bond is an important step in synthetic organic chemistry [1]. Acetophenone, a product of ethylbenzene oxidation, is widely used in the manufacturing of pharmaceuticals, resins, fragrance and as a solvent for cellulose ethers [2]. It is traditionally produced by acylation of benzene via the Friedel-Crafts reaction, which generates a large amount of waste [3]. Replacing the Friedel-Crafts process by selective oxidation of styrene or ethylbenzene has been explored extensively using stoichiometric oxidants [4–11], but the green usage of molecular oxygen is still a challenge [1,9,12–14].

Various transition metals including Co, Mn, Cr and Fe have been examined as catalysts for ethylbenzene oxidation [5,6,15–25]. Bhoware et al. synthesized 1.0–2.0 wt% cobalt-containing hexagonal mesoporous molecular sieves (Co/HMS) and observed high activity in ethylbenzene oxidation using tert-butyl hydroperoxide as the oxidant [20]. For the usage of molecular oxygen as an oxidant, the reaction was carried out under severe reaction conditions [19,23,26]. High pressure of 1 MPa O_2 was required, when Ma et al. used 1.6 wt% Co/SBA-15 to catalyze the ethylbenzene oxidation [19]. Lin et al. prepared cobalt nanoparticles encapsulated into graphitic nitrogen-doped carbon nanotubes

(Co@GCNs) which showed efficient oxidation of ethylbenzene to acetophenone under 0.8 MPa O_2 pressure [23]. Thus, designing a green catalyst as a highly dispersed Co species through the encapsulating treatment might be efficient for selective oxidation with atmospheric pressure O_2 . However, it is difficult to find a suitable support or ligand not only for metal species anchoring but also for the synergistic promotion.

There has been no report to date of using solubilized transition metal clusters as catalysts for ethylbenzene oxidation. But there are many interesting reports of catalysis by quantum-sized metal clusters [27–30]. Corma et al. found that gold clusters of low atomicity had extraordinarily high activity in thiophenol reaction with molecular oxygen [27]. Oliver-Meseguer et al. reported that small gold clusters (3 to 10 atoms) catalyzed the ester-assisted hydration of alkynes with a high turnover frequency (TOF) of about 10^5 per hour [28]. Liu et al. synthesized atomically dispersed palladium catalysts that were about 55 times more active than the commercial Pd catalysts in the hydrogenation of aldehydes [29]. Unusual catalytic activities were found for metal nanoclusters catalysis in the oxidation reactions [27,31–34]. Previously, we found that leached Au from Au/SiO_2 *in situ* forms suitable Au_{7-8} clusters on which the superoxo species is formed to oxidize

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